

AlgoLab Summer School 2026
Final exam
June 19, 2026

This is a three-hour multiple-choice exam. For each question, select exactly one answer and provide a brief justification for your choice. Your justification may take the form of a short sentence, a mathematical equation, a graph, a diagram, or a few explanatory lines, as appropriate, and should show the reasoning behind your answer, but it does not need to be long. You are allowed to use your course notes during the exam. All other resources, including web searches and online AI tools, are not permitted. Good luck!

Question 1 (Adam Bene Watts)

For any n -bit classical function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let U_f be the $(n + 1)$ -qubit unitary which computes that function in an ancilla register, so

$$U_f |x\rangle |0\rangle = |x\rangle |f(x)\rangle$$

for each n -qubit computational basis state $|x\rangle$. On average, the number of gates from the Clifford+ T gate set ($\{H, \text{CNOT}, Z, S, T\}$) required to implement the unitary U_f for an n -bit function f :

- (a) is constant.
- (b) grows polynomially with n .
- (c) grows exponentially with n .
- (d) trick question: this gate set cannot implement the unitary U_f .

Correct answer: C.

Question 2 (Arthur Mehta)

Which of the following chains of containments is known to be correct?

- (a) $P \subseteq BQP \subseteq BPP \subseteq QMA$, $P \subseteq NP \subseteq MA \subseteq QMA$, $IP \subseteq QMA$.
- (b) $P \subseteq BPP \subseteq BQP \subseteq QMA$, $P \subseteq MA \subseteq NP \subseteq QMA$, $IP \subseteq NP$.
- (c) $P \subseteq BPP \subseteq BQP$, $NP \subseteq MA \subseteq QMA$, $IP \subseteq MIP \subseteq MIP^*$.
- (d) $P \subseteq BPP \subseteq BQP \subseteq QMA$, $P \subseteq NP \subseteq MA \subseteq QMA$, $MIP^* \subseteq QMA$.

Correct answer: C.

Question 3 (Roeland Wiersema)

After absorbing all surrounding gates into an effective state $|\psi\rangle$ and an effective observable O , the VQE cost with respect to a single parameter θ_j can be written as

$$C(\theta_j) = \langle \psi | U_j^\dagger(\theta_j) O U_j(\theta_j) | \psi \rangle, \quad (1)$$

where $U_j(\theta_j) = e^{-i\theta_j P_j/2}$, $P_j^2 = I$. Define $A(\theta_j) \equiv U_j^\dagger(\theta_j) O U_j(\theta_j)$.

Differentiating gives

$$\partial_{\theta_j} C(\theta_j) = \frac{i}{2} \langle \psi | [P_j, A(\theta_j)] | \psi \rangle. \quad (2)$$

Using $P_j^2 = I$, one can show

$$e^{i\pi P_j/4} = \frac{1}{\sqrt{2}}(I + iP_j), \quad e^{-i\pi P_j/4} = \frac{1}{\sqrt{2}}(I - iP_j). \quad (3)$$

Hence,

$$e^{i\pi P_j/4} A(\theta_j) e^{-i\pi P_j/4} - e^{-i\pi P_j/4} A(\theta_j) e^{i\pi P_j/4} = i[P_j, A(\theta_j)]. \quad (4)$$

Which expression gives the exact derivative $\partial_{\theta_j} C(\theta_j)$?

$$(a) \quad \partial_{\theta_j} C(\theta_j) = \frac{1}{2} \left[C\left(\theta_j + \frac{\pi}{4}\right) - C\left(\theta_j - \frac{\pi}{4}\right) \right].$$

$$(b) \quad \partial_{\theta_j} C(\theta_j) = C\left(\theta_j + \frac{\pi}{4}\right) - C\left(\theta_j - \frac{\pi}{4}\right).$$

$$(c) \quad \partial_{\theta_j} C(\theta_j) = \frac{1}{2} \left[C\left(\theta_j + \frac{\pi}{2}\right) - C\left(\theta_j - \frac{\pi}{2}\right) \right].$$

$$(d) \quad \partial_{\theta_j} C(\theta_j) = C(\theta_j + \pi) - C(\theta_j - \pi).$$

Correct answer: C.

Question 4 (Abhijeet Alase)

Approaches for implementing functions of Hermitian matrices are useful for which of the following applications?

- (a) Hamiltonian simulation
- (b) Quantum linear system solvers
- (c) Quantum semidefinite program solvers
- (d) All of the above and many more

Correct answer: D.

Question 5 (Ulrike Stege)

When implementing a problem in unconstrained binary optimization (UBO) form for the use of standard QAOA as proposed by Farhi et al. (2014), one

- (a) implements the cost hamiltonian as quantum circuit using 1-qubit Pauli- Z gates
- (b) encodes the UBO into the mixer hamiltonian
- (c) implements the matrix exponential of the cost hamiltonian as quantum circuit using Z -rotation gates
- (d) is guaranteed that the optimum solution is found by 1-layer of the QAOA

Correct answer: C.

Question 6 (Chloé Gauvin-Ndiaye)

Which of the following statements about the Hubbard model is true?

- (a) There are some parameter regimes in which the Hubbard model can be solved exactly using classical approaches.
- (b) The Hubbard model offers a realistic description of all strongly-correlated electron systems.
- (c) The Hubbard model is defined using the second quantization formalism because we need to quantize the fermions twice to get the proper sign of the ground state energy.
- (d) The interacting part of the Hubbard model is quadratic in terms of creation and annihilation operators, meaning proportional to $c^\dagger c$.

Correct answer: A.

Question 7 (Konstantinos Meichanetzidis)

When establishing the threshold for quantum advantage in estimating the Jones polynomial, the methodology moves beyond purely theoretical asymptotic complexity. Based on the framework presented, which of the following best describes how the crossover point for quantum advantage is calculated?

- (a) By calculating the exact knot crossing number where the classical tensor network contraction scaling transitions from subexponential to strictly exponential time, independent of the target accuracy.
- (b) By demonstrating that the DQC1-complete Markov closure of any braid cannot be approximated by any classical method under a time budget of 1 hour on one GPU.
- (c) By finding the problem size where the quantum time-to-solution or energy consumption drops below that of the best tried approximate classical algorithm, evaluated at a fixed additive error and constrained by explicit hardware parameters (e.g., classical FLOPS vs. quantum gate error and layer time).
- (d) By comparing the quantum execution time of the quantum algorithm against its exact classical state-vector simulation running on an exascale supercomputer.

Correct answer: C.

Question 8 (Sisi Zhou)

In phase estimation where a single qubit is evolving under a $e^{-iZ\theta}$ rotation and θ is to be estimated, which of the following is an optimal input state to maximize the sensitivity?

- (a) $(|0\rangle + |1\rangle)/\sqrt{2}$
- (b) $(|0\rangle + \sqrt{3}|1\rangle)/2$
- (c) $\cos \frac{\pi}{8} |0\rangle + \sin \frac{\pi}{8} |1\rangle$
- (d) none of the above.

Correct answer: A.